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The Number They Didn't Give

Google's 13 billion, the Loviisa contract, and what it costs a Finnish house heated by electricity

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A Finnish household pays 18.0 cents per kilowatt-hour for electricity once distribution and taxes are included. A detached house heated by electricity uses about 11,000 kilowatt-hours a year, and the competitive energy share of its bill came to 762 euros last season. Finland's electricity has been among the cheapest in Europe.

That was already changing before Wednesday. Measured consumption is growing by roughly 400 megawatts a year without a single data centre, the largest single driver being electric boilers in district heating networks. Production is being built to follow, but what gets built is wind, and wind produces when the wind blows. Without Google, without any new announcement at all, the same house sees its annual bill rise by about 226 euros by 2030.

And there is a second question, which is not about price. In the 2025/26 season, at the tightest hour of the year, Finland's entire 9,535 megawatts of installed wind capacity produced 162 megawatts. Adequacy is not settled in annual energy but in the hours of January and February when it is cold and still. There were well over a hundred of them last season, and more wind is not the answer to them.

This text was written on the evening of the news day and extended afterwards. Part I covers what happened and what was said, Part II what follows from it without qualification, Part III the calculations, the data and the method. The caveats are in Part III and the change log is in Section 26.

PART I

The Day

1 What happened

On Wednesday 9 September 2026, Google announced it would invest at least 13 billion euros in Finland during 2027–2028. The money goes to expanding the Hamina data centre and to new centres in Kajaani, Vaala and Muhos. It is the company's largest single investment in Europe.

On the same day Fortum and Google signed a 22-year power purchase agreement. It covers up to half of the capacity of the Loviisa nuclear power plant in 2030–2049 and begins with a smaller volume in 2028.

Let me start with one confusion, because it went wrong in public and because it is the heart of the matter.

The agreement does not concern Olkiluoto 3. It concerns Loviisa. Olkiluoto 3 is 1,600 megawatts of new capacity added to the system in 2023. Loviisa is 1,014 megawatts of old capacity that was on its way out. According to Fortum's own statement, the plant would not be able to produce electricity after 2030 without substantial lifetime-extension investments, and 80 per cent of those projects and 700 million euros were still without an investment decision the day before the announcement.

The news therefore does not report an increase in electricity. It reports that a decrease will not happen. That is a significant piece of news, but it is a different piece of news from the one that was read.

Table 1. What the package contains

Item	Amount	Timing
Loviisa capacity	2 × 507 MW	in service since 1977 and 1981
Loviisa annual output	approx. 8 TWh	approx. 10 % of Finland's electricity
Share covered by the agreement	up to 50 %	2030–2049
Initial phase	smaller volume	2028–2029
Uprate under way	38 MW	complete in 2028
Further uprate enabled by the deal	10 MW	
Battery storage in Kajaani	94 MW	
Lifetime extension programme	approx. €1 bn	to 2050
of which without investment decision	80 % of projects, €700 m	
Google's investment	at least €13 bn	2027–2028
Power demand of Google's data centres	**not disclosed**	

The last row is the title of this analysis.

2 What was said

The news day produced more statements than numbers. I go through them by asking of each one which question it actually answers. There are three questions, not one. **A:** is there enough energy on an annual basis. **B:** will the winter average price rise. **C:** what happens at the critical hour, and what does the bill do. Those who are worried are asking C. Those who answered, answered A and B.

Table 2. The day's claims and which question they answer

Claim and speaker	Question	Assessment
Prices will not rise because of the investment (Orpo)	B	holds as an average
There is enough electricity, no cause for worry (Orpo)	A	holds as energy
Google carries its share of flexibility (Orpo)	C	refuted the same day
Bring your own power (Porat, Google)	A and C	describes finance, not capacity
Data centres barely flex (Järvinen, Google)	C	the day's key sentence
Continued operation stabilises prices (Rauramo, Fortum)	B and C	holds as a counterfactual
The average price should not rise (Leskelä, ET)	B	holds, with its caveats
Flexibility settles the spike (Honkapuro, LUT)	C	partly; see Section 15
Google wins, others lose (Vehviläinen, Aalto)	C	depends on a race

2.1 Three statements from one day that do not fit together

Prime Minister Petteri Orpo said at Google's event that he was convinced there is enough electricity and that Finns should not be worried. As grounds he said the government had been assured of this. He also said that Google carries its own share of adding production and of flexibility.

A few hours later Google's country manager for Finland, Antti Järvinen, said from the same event that data centres do not flex during price spikes, and he measured the amount of flexibility: it is *single percentage points*. He then pointed to the Kajaani battery as the source of flexibility.

In the same context the Prime Minister mentioned two other things. The government is preparing a price cap for households in case of a crisis, and the Meri-Pori coal-fired plant must be held in reserve.

These three point in different directions. The first says there is no problem with the critical hour. The two others are measures prepared only if there is a problem with the critical hour. A price cap is not prepared for an average, and a coal plant is not held in reserve on account of annual energy.

This is not a matter of dishonesty. The answer was given to question B while the preparedness is aimed at question C, and neither is said out loud.

On flexibility the situation is clearer. Single percentage points of a load in the order of a thousand megawatts is tens of megawatts. The claim that Google carries the responsibility for flexibility does not hold in the unit in which flexibility is measured — and it was refuted by the company's own representative on the same day.

2.2 Bring your own power

Google's global investment director Ruth Porat described the company's principle as *bring your own power*: the company adds production capacity itself rather than merely consuming what exists.

The principle is sound. The contents of this package do not match it. What is new is the 38 megawatts of the Loviisa uprate and the further 10 megawatts the agreement enables. The rest is a purchase agreement on an existing plant and a battery, which does not produce but shifts. Fortum's 629 megawatts of new onshore wind is a Fortum project whose construction was under way before the agreement.

A more accurate formulation would be *keep your own power*. It would not be a more modest claim — preventing a closure is by far the largest single effect in this package — but it would be correct.

2.3 The number they didn't give

Google did not disclose the electricity consumption of its data centres. Järvinen also did not break down the investment by location or give a schedule.

Two outside estimates were offered the same day. Iivo Vehviläinen, professor of practice in energy economics at Aalto University, estimated that the new demand corresponds to roughly one large nuclear unit. Samuli Honkapuro, professor of electricity markets at LUT University, called the addition significant relative to Finland's average hourly consumption, and noted that full consumption landing on top of a price spike makes those spikes worse — but that with flexibility the extra load would fit into the system well.

One large nuclear unit means 1,400–1,600 megawatts in Finnish conditions.

The entire outcome of this analysis depends on that number, and it is presented in Section 21.

2.4 The price cap, and why it does not answer this

The Prime Minister's mention of a household price cap has recurred in the discussion since. Four counter-arguments have been made in public that are better than my own, and I record them here because they are right.

A cap means — or should mean — a situation in which Europe is in an electricity crisis and emergency conditions are jointly declared. On ordinary winter cold days no crisis is declared, so the cap does not apply in precisely the hours for which it is demanded. The three others concern what would happen if a cap were nonetheless imposed: consumption would continue unchanged in the middle of a shortage, producers would lose the incentive to generate exactly when it is needed most, and the question of who pays for the cap is left open.

The arguments are those of Anni Lassila (HS Visio) and Jussi Kärki (Talouselämä). I had written only that a cap treats a symptom rather than a cause. That is thin beside these, and I replace it with them.

2.5 Two legislative initiatives and one support scheme that fell

MP Timo Harakka (Social Democrats) proposed a mechanism in the Electricity Market Act that would guarantee households some of the cheapest electricity in Europe in future, along with an obligation for large consumers to use their backup power when required. MP Heikki Vestman (National Coalition) has tabled a bill on connection conditions for data centres, signed by more than a hundred members, and has demanded a capacity mechanism.

A little over a week before Google's announcement, the government decided in its budget session to drop a planned new support scheme for data centres worth at most 30 million euros a year. The data centres' electricity tax relief had ended at the start of July, when the rate rose from 0.05 to 2.24 cents per kilowatt-hour.

Orpo's reasoning for dropping the support was that it would have remained small relative to the size of the investments, and that interest in Finland is enormous.

Eight days later Google announced 13 billion. The tax rose 45-fold, the compensation fell, and the investment came anyway.

PART II

Conclusions

In the near term (2026–2029) this will not show in your bill. Google's load does not exist yet, and data centres start smaller than their permit applications state. Norwegian reservoirs stood at 63.8 per cent in week 36, against a thirty-one-year median of 82.0 for the same week, with only two lower years on record. That is risk information, not a forecast — Section 11 shows that reservoir levels predict the winter price less well than I have previously claimed. But if the winter turns out expensive, the reason is not Google.

In the long term (2030–) the agreement is a gain for the consumer. Everything turns on one number Google did not disclose: the break-even runs at 1,014 megawatts, exactly Loviisa's capacity, because the deal is a wash when Google's load equals the capacity it saves. The best independent estimate of that load is 300–500 megawatts, a third to a half of the threshold. On that estimate the house's bill *rises 109–149 euros a year less* than in the world where no agreement was made and Loviisa closed in 2030. The bill does not fall in any scenario — it rises in all of them, and the agreement merely slows the rise.

And the bill rises regardless. Without a single data centre, on ordinary demand growth alone, the wholesale annual bill of a detached house heated by electricity rises by about 226 euros, or 30 per cent, by 2030. Google's own share on top of that is 57–97 euros a year. The whole data centre wave would be 989 euros a year.

The problem is not Google. Its agreement is the only data centre project in Finland that brings a counterweight with it. The others do not. Projects with signed connection agreements amount to nearly five gigawatts, and if they are built and no production is added, the same house pays 1,064 euros a year more.

Who pays. In the 2025/26 season a detached house heated by electricity paid 69.26 euros per megawatt-hour, against an arithmetic average of 56.47 and 62.03 for industry. Households account for 34.8 per cent of consumption but 39.0 per cent of consumption during the most expensive five per cent of hours. Industry dodges the expensive hour. A household does not, because the cold is the cause of both its consumption and the price.

Scarcity is a winter phenomenon, and that is why wind is not enough. In February 2026 consumption was at its highest and the wind share at its lowest, and the price was 137 euros per megawatt-hour. In March consumption equalled December's but wind supplied over a third of production, and the price was 28. During the most expensive hours wind produces a third of its usual output. Doubling wind capacity therefore restores the annual average price but leaves the most expensive hours and February still elevated — while the same amount of weather-secure capacity restores all three exactly.

Imports from neighbours do not help, because they are already in use. During the most expensive one per cent of hours, 2,674 megawatts were imported from Sweden and the link was more than 90 per cent full in one hour out of three. At the same time Finland's price was 202 euros per megawatt-hour above central Sweden's. The price gap is proof that the link is full. When it is cold and still here, it is cold and still next door, and the neighbour has nothing to sell.

Demand grows without data centres too. Consumption in the distribution networks grew 7.0 per cent in a single year, or 3.47 terawatt-hours, equivalent to 397 megawatts of flat load. The largest single contributor is electric boilers in district heating, whose consumption rose 37 per cent. Google's 1,600 megawatts would therefore be four years of ordinary growth arriving at once. Data centres are not the cause of the problem; they are the messenger.

And what I was about to say wrong myself. I had thought the problem was the 278 cold and still hours and that demand flexibility was the cheap fix. For the bill it is not. When I calculated it, removing the load from the most expensive five per cent of hours cut five per cent off the effect. The bill is not made in the tail. It is made across all 8,760 hours.

What has to happen. Only new energy fixes the bill, and roughly megawatt for megawatt. But not all megawatts are equal: doubling wind capacity restores the annual average price, while weather-secure capacity restores February and the most expensive hours as well. Flexibility and backup power are necessary, but they answer adequacy rather than price, and they should not be sold to the reader as a price solution. And because demand grows by some 400 megawatts a year in any case, building has to continue after the data centres are built.

YARDSTICK

According to Statistics Finland, the all-in price of electricity for a household is **18.0 cents per kilowatt-hour** as a twelve-month average, of which energy is 5.9, distribution 5.5 and taxes 5.7. The competitive energy component is **one third of the bill**. Every figure in this analysis concerns that third — not the whole bill.

PART III

Background, Data and Method

3 Concepts

Energy and capacity. Energy is terawatt-hours per year, capacity is megawatts at one moment. The annual balance can be in surplus while the system is in deficit during a single January hour. Adequacy is settled in the latter unit.

Firm capacity. Capacity that is available when it is called for, regardless of the weather. Nuclear, hydro and condensing capacity are firm. Wind is not: a turbine's output is the cube of wind speed, so halving the speed removes 87 per cent of production.

Flat load. Consumption that does not depend on outdoor temperature. Heating load disappears in mild weather; a server hall does not. Flat load does not change the shape of the peak but it raises the floor on which the peak is built.

Residual load. Consumption minus wind production. It is the variable that explains price best, because it states how much dispatchable production and import are required.

PPA. A long-term power purchase agreement. It is a financial instrument, not a transmission line. Loviisa's electricity enters the Finnish system regardless of the agreement; the agreement only determines whose balance it is booked to and at what price.

4 Data

Fingrid open data. Dataset 358 gives consumption in Finland's distribution networks by hour and customer type (consumer and business), and dataset 360 the same by user group under the Statistics Finland classification, with households broken down by heating method. Both are based on Datahub metering data and begin on 1 August 2023. In addition datasets 124 consumption, 75 wind, 188 nuclear, 194 net imports, 201 district heating CHP, and 60 and 61 cross-border flows to Sweden.

Price. Finland's area price by hour from the Sähkötin service, 27,048 hours for the simulation and 1 January 2015 to 31 August 2026 for the reservoir comparison in Section 11, a total of 102,264 hourly observations. Cross-checked against the energy-charts series: mean deviation 0.000 euros per megawatt-hour, largest single deviation 2.99.

Reservoirs. NVE's weekly statistics 1995–2026, Norway's five price areas weighted by capacity.

Temperature. ERA5 reanalysis via the Open-Meteo archive.

Other. Annual statistics from Energiateollisuus and Statistics Finland, Fingrid's half-year report, government tax decisions, and the report *Datakeskusten faktat ja fiktiot* by Pervilä and Kangasharju (University of Helsinki, 20 July 2026).

The panel combining all hourly series covers 27,002 hours from 1 August 2023 to 31 August 2026.

5 Method

The price effect is calculated by fitting price as a function of residual load and simulating the addition of a flat load.

The fit is a k-nearest-neighbour regression: for each grid point the 400 nearest hours by residual load are taken and their mean price computed. The curve is made monotone by cumulative maximum, because a supply curve does not fall. Outside the observed range it is extrapolated using the slope of the final segment.

The simulation is applied as a difference, not as a replacement. For each hour $\Delta p = f(\mathrm{RL} + X) - f(\mathrm{RL})$ is computed and the new price is the realised price plus Δp, floored at zero. This preserves each hour's own variation and uses only the estimated slope from the model.

Each user group's bill is computed hour by hour on its own consumption, not on an average. This matters, because the groups consume at different times.

What the model assumes. Supply stays as it is. Nothing is built, nothing is retired, and the other Nordic price areas behave as they do in the data. The result is therefore not a forecast but an upper bound for the case where load arrives and no production is added. That is a deliberate choice: the gap between this bound and the built world is the policy question, and its size is measured in Sections 6 and 15.

Validation. The share of extrapolation was checked. At an added load of 1,600 megawatts, 2.6 per cent of hours exceed the observed maximum residual load, and those hours account for 0.1 per cent of the total price effect. At 2,500 megawatts the figures are 6.4 and 4.3 per cent. The result is therefore not an artefact of extrapolation at additions below 2,000 megawatts.

6 How seriously to take the five-gigawatt figure

Fingrid's chief executive Asta Sihvonen-Punkka said at the end of August that the company expects electricity consumption to grow by as much as 40 per cent over the next five years. The view rests on signed connection agreements amounting to some 5,000 megawatts. That figure underpins the largest scenarios in this analysis, so its reliability deserves to be opened up.

A connection agreement is a stronger signal than a connection enquiry, because it is not made without an intention to build. It is still not built capacity.

The Confederation of Finnish Industries has published a realisation ladder for investment projects that applies here directly: of projects at the preliminary study stage some five per cent are realised, of those at the planning stage about one in ten, of those with a completion date about 40 per cent, and after an investment decision realisation is close to certain.

A second checkpoint is blunter. There is at present exactly one data centre in Finland above one hundred megawatts — Google's Hamina. It grew to about 80 megawatts over thirteen years. The entire current data centre fleet is around 300 megawatts.

Three other operators have publicly named projects: Microsoft is building three sites in Uusimaa with a published capacity of 615 megawatts, and DayOne is building 150 megawatts in Kouvola and 130 in Lahti for TikTok. That is 895 megawatts of named non-Google capacity.

Two things follow. The five-gigawatt scenario should not be read as a forecast — it is the sum of connection agreements, and the historical realisation rate in every investment class is below one hundred per cent. But it should not be dismissed either, because it is the strongest information available and because the transmission grid must be dimensioned on what has been agreed, not on what materialises.

In this analysis the largest scenarios are used as an upper bound, and the row that carries the most weight is Google's own load.

7 Who pays what: measured, not estimated

Public discussion uses the arithmetic average price. Nobody pays the arithmetic average price. Season 2025/26, July to June, 8,760 hours:

Table 3. What each user group actually paid, season 2025/26
Volume-weighted wholesale price. Arithmetic average 56.47 €/MWh.

Group	TWh	Metering points	€/MWh	Difference
Holiday homes	1.61	342,027	71.99	+15.52
Detached house, electric heating	11.44	1,039,263	**69.26**	+12.79
Detached house, other heating	2.21	325,157	67.70	+11.23
Residential properties	3.12	178,074	66.53	+10.06
Agriculture	1.92	97,797	65.70	+9.23
Apartments	3.13	1,473,112	62.85	+6.38
Services	12.45	173,723	62.25	+5.79
Industry	9.83	31,505	**62.03**	+5.57
Utilities and energy supply	3.34	38,015	55.40	−1.07

A detached house heated by electricity paid 7.23 euros per megawatt-hour more than industry. Not because it has a worse contract, but because it consumes at the wrong time.

The exposure can be measured directly. The most expensive five per cent of hours, threshold 165 euros per megawatt-hour:

Table 4. Exposure to expensive hours, season 2025/26

Group	Share of consumption	Share of most expensive	Factor
Holiday homes	3.0 %	3.8 %	1.257
Detached house, electric heating	21.7 %	24.8 %	1.147
Detached house, other heating	4.2 %	4.6 %	1.106
Apartments	5.9 %	5.7 %	0.954
Services	23.6 %	22.4 %	0.949
Industry	18.6 %	17.2 %	0.925
Households in total	34.8 %	39.0 %	1.119

The wholesale bill of households over the season came to 1,254 million euros, and that of all distribution network consumption to 3,388 million.

8 The critical hour

The consumption record of the 2025/26 season was set on 8 January 2026. Fingrid has confirmed the figures: a quarter-hour average of 15,279 megawatts, a highest reading of 15,553, and a domestic production record of 14,327. The previous record was 15,105 megawatts, from 2016.

The consumption peak is not, however, the tightest moment. The tightest is cold and still at the same time. There were 278 such hours, 7.7 per cent of winter hours, with imports averaging 2,908 megawatts and peaking at 3,233 — the entire added capacity of the Aurora Line was absorbed into them.

The anatomy of the expensive hour tells what is going on. Temperature averaged -11.4 degrees against a winter mean of -1.8, but the wind share was 4.6 per cent against a winter mean of 24.6. The cold is ordinary winter cold. The stillness is not ordinary.

And the expensive hour has become Finnish. Of Finland's most expensive hours, 88 per cent were also expensive in Sweden's SE3 area in the seasons 2020/21–2022/23; in 2025/26 only 36 per cent. When the price separates from the neighbour's, it is set by domestic scarcity.

9 Why scarcity is a winter phenomenon

The price spike and the capacity shortage are not scattered across the year. They are in the same place, and the monthly profile shows why.

Table 5. Season 2025/26 by month, hourly averages

Month	Consumption MW	Wind MW	Wind share %	Residual load MW	Price €/MWh
November	10,582	2,952	27.9	7,629	47.9
December	11,055	3,233	29.2	7,822	36.1
January	12,892	2,160	16.8	10,732	**117.2**
February	**13,020**	**1,795**	**13.8**	**11,225**	**137.2**
March	11,055	3,947	35.7	7,107	27.7
July	8,128	1,185	14.6	6,942	24.3

March and December are the instructive pair. Consumption in both was 11,055 megawatts, exactly the same. In March there were 3,947 megawatts of wind and the price was 27.7 euros; in December 3,233 and the price 36.1. And in February, when consumption was only 18 per cent above March's, there was less than half the wind and the price was five times higher.

Consumption alone therefore explains nothing. What matters is that two things coincide: consumption is high and wind output is low. In winter months the correlation between consumption and wind share is -0.189 — they move in opposite directions precisely when it matters.

The same shows in the simulation. Half of the price effect of an added 1,600 megawatts arises in November to March, even though those months are 41 per cent of the hours. In winter the average price increase is 33.0 euros per megawatt-hour and in the other months 23.1.

9.1 What this means for building more wind

Here the annual balance and the single hour diverge most sharply.

During the most expensive five per cent of hours of the 2025/26 season, wind production averaged 783 megawatts against a whole-season mean of 2,389. Wind delivers a third of its usual output at the expensive hour. It follows that doubling capacity adds 2,389 megawatts to the annual average but only 783 to the expensive hour.

Table 6. The same measure on three different yardsticks, season 2025/26
Price in euros per megawatt-hour

Scenario	Whole year	Most expensive 5 %	February
Realised	56.51	232.8	137.2
+1,600 MW, nothing done	83.70	255.1	176.2
+1,600 MW and wind capacity ×1.5	66.25	249.8	156.8
+1,600 MW and wind capacity ×2.0	**56.63**	241.2	140.9
+1,600 MW and 800 MW weather-secure	70.13	246.9	160.0
+1,600 MW and 1,600 MW weather-secure	**56.51**	**232.8**	**137.2**

Doubling wind capacity restores the annual average price to its starting level but leaves the most expensive hours 8.4 euros and February 3.7 euros higher. Weather-secure capacity restores all three exactly.

This is the heart of the matter in a single table. Wind fixes the year. Only weather-secure capacity fixes the hour. And because a household consumes proportionally most precisely when the wind is not blowing, fixing the year is not enough for it.

10 Would imports from the neighbours help

A recurring idea in public discussion is that the common Nordic market handles scarcity: when Finland is expensive, electricity is imported from Sweden. I measured it.

In the 2025/26 season the largest observed flow from Sweden to Finland was 3,298 megawatts. That is the practical ceiling against which everything else can be compared.

Table 7. Imports from Sweden and the price gap to SE3, season 2025/26

Hours	Import avg MW	Link over 90 % full	FI price €/MWh	FI − SE3 €/MWh
All 8,754 hours	1,271	4.2 %	56.5	−5.4
Most expensive 5 %	2,284	21.7 %	232.8	+80.3
Most expensive 1 %	2,674	32.2 %	358.7	+202.0
Still and heavily loaded hours	2,795	32.0 %	163	+42

Still and heavily loaded hours means hours in which the wind share was below ten per cent and consumption above 11,000 megawatts. There were 709 of them.

The figures say three things.

Imports are not a reserve; they are already in use. In an ordinary hour 1,271 megawatts are imported from Sweden and the link is near full in four per cent of hours. In the most expensive one per cent, 2,674 megawatts are imported and the link is near full in one hour out of three. The market uses the link first, and only then does the price rise. By the time the price is high, the help has already been taken.

The price separates from the neighbour's precisely when help would be needed. Over the whole season Finland was on average 5.4 euros cheaper than central Sweden. In the most expensive one per cent Finland was 202 euros more expensive. The price gap is the signature of a full link: if capacity were free, prices would converge.

And the cause is shared weather. Cold and still weather does not stop at the Finnish border. When it is cold here it is cold in Sweden too, and when the wind fails here it fails there. The neighbour has the same problem at the same time and nothing spare to sell.

New transmission capacity is therefore not the answer to this hour. The Aurora Line was completed, and its entire added capacity was absorbed immediately into the cold and still hours — the import maximum rose from 2,753 megawatts in earlier winters to 3,233, and the price spike did not disappear.

11 Hydropower, and a correction to my own published analysis

In my analysis *Kallis talvi, kallis tunti* (20 August 2026) I argued that the winter price level is set by Norway's hydrological situation, and the evidence was a correlation of -0.73 between the November reservoir level and the winter average price. The sample was six winters, 2020/21–2025/26.

I extended the sample to eleven winters, because the price series is available back to 2015.

Table 8. Norwegian reservoirs in November and the winter average price in Finland

Winter	Reservoir level in November	Winter average price €/MWh
2015/16	87.5 %	29.9
2016/17	73.5 %	34.8
2017/18	83.9 %	38.3
2018/19	78.6 %	49.0
2019/20	75.9 %	31.2
2020/21	94.3 %	42.5
2021/22	71.1 %	111.5
2022/23	79.8 %	135.5
2023/24	77.8 %	72.9
2024/25	86.2 %	46.3
2025/26	80.0 %	72.1

The correlation depends decisively on which winters are included.

Table 9. The same relationship on four different samples

Sample	n	Correlation
2020/21–2025/26, as previously published	6	-0.705
2015/16–2025/26, full series	11	-0.364
2015/16–2025/26 excluding the crisis winters 21/22 and 22/23	9	-0.159
2015/16–2020/21, before the crisis	6	$+0.153$

The six-winter sample reproduces, so my earlier calculation was not erroneous. It was short. When the sample is lengthened the relationship weakens to a third and effectively disappears if the two energy-crisis winters are excluded. In the six winters before the crisis the sign is even wrong.

The explanation is obvious in hindsight. The winters 2021/22 and 2022/23 were expensive, and the cause was European gas, not Norwegian water. They happened to fall in years when reservoir levels were somewhat below average, and in a sample of six observations two such points determine the slope.

What is corrected and what is not. The claim that *the November reservoir level predicts the winter average price* does not survive a longer sample, and it must be corrected. The claim that *hydropower is a central price setter in the Nordic market* is a different claim and is not overturned by this: hydro is often the production form that sets the price, and its scarcity shows in the price. What proved weak is the predictive power of one annual indicator.

The practical consequence for this analysis: the coming winter's price should not be forecast from reservoir levels. In week 36 of 2026 Norway's reservoirs stood at 63.8 per cent, which is in the lowest decile of the thirty-one-year distribution, and that is worth telling the reader — but it is risk information, not a forecast.

12 Demand grows without data centres too

The whole data centre discussion is conducted as though other consumption stood still. It does not.

The Datahub data allow measurement of how much consumption in the distribution networks grew between the two most recent full seasons. Data centres are not a separate item in these groups, and the largest of them connect directly to the transmission grid.

Table 10. Distribution network consumption 2024/25 → 2025/26, TWh

Group	2024/25	2025/26	Change	Metering points
Utilities and energy supply	2.43	3.34	+37.2 %	+1.4 %
Electric vehicle charging	0.10	0.16	+53.7 %	+36.8 %
Construction	0.22	0.25	+12.6 %	+1.4 %
Detached house, electric heating	10.62	11.44	+7.8 %	+0.7 %
Services	11.73	12.45	+6.2 %	+0.2 %
Industry	9.31	9.83	+5.6 %	−1.6 %
Apartments	3.01	3.13	+4.0 %	+0.5 %
Total	**49.32**	**52.79**	**+7.0 %**	

Consumption grew by 3.47 terawatt-hours in one year. Converted to flat load that is 397 megawatts.

The largest single contributor is utilities and energy supply, whose consumption rose 37 per cent while the number of metering points grew 1.4 per cent. The cause is electric boilers in district heating networks. According to Fingrid, the combined capacity of electric boiler projects in operation and under construction exceeded three gigawatts in June 2026. That is the same order of magnitude as the capacity of data centres with signed connection agreements, and it is not discussed at all.

Electric vehicle charging grew 54 per cent and its metering points 37. In terawatt-hours it is still small, but it is the fastest growth in the data.

A caveat that must be made. The 2025/26 season was colder than 2024/25, and part of the growth is weather rather than structure. This applies particularly to detached houses with electric heating. The 37 per cent growth in electric boilers is too large to be explained by weather alone in a group whose metering point count did not grow, and Fingrid's three-gigawatt figure confirms it independently. The growth in vehicle charging is metering point growth, not weather.

What this does to the whole calculation. Four hundred megawatts a year is ordinary growth without a single data centre. Google's 1,600 megawatts is therefore four years of ordinary demand growth in size — but it arrives at once and is announced as a single news item, while electric boilers and heat pumps arrive unnoticed.

And it turns one thing on its head. If Finland does not build new production, the price rises without data centres too. Data centres are not the cause of the problem but its messenger, and blocking them does not remove the fact that more electricity is needed in any case.

13 What the agreement adds to the critical hour

Table 11. New and secured capacity at the critical hour

Item	Capacity	Type
Loviisa uprate, under way	38 MW	firm, new
Further uprate enabled by the agreement	10 MW	firm, new
Kajaani battery storage	94 MW	firm, limited in energy
Fortum's 629 MW of new wind, at the expensive hour's 4.6 %	29 MW	weather-dependent
New at the critical hour	approx. 170 MW	
Existing capacity secured by the agreement	1,014 MW	firm, old

The ratio is about one to six. On the battery one clarification is needed: its capacity is 94 megawatts but its energy is limited, 188 megawatt-hours as a two-hour battery. That covers a five-hundred-megawatt load for 22 minutes. Still periods last days. A battery is the right tool for the quarter-hour, not for the week.

14 The price effect: my own calculation

The simulation result for the 2025/26 season, with supply held fixed.

Table 12. Adding flat load, season 2025/26 (realised average price 56.50 €/MWh)

Added load	Average price	Change	Households	Elec.-heated house
+500 MW	65.11	+15 %	+178 m€/yr	+108 €/yr
+1,000 MW	73.45	+30 %	+345 m€/yr	+209 €/yr
+1,600 MW	83.70	+48 %	+546 m€/yr	+331 €/yr
+2,500 MW	100.56	+78 %	+870 m€/yr	+525 €/yr
+5,000 MW	156.62	+177 %	+1,790 m€/yr	+1,064 €/yr

The same run for the 2024/25 season, which was mild and windy, gives +81 per cent for 1,600 megawatts and 297 euros a year per house. The order of magnitude of the effect therefore does not depend on whether it lands in an expensive or a cheap year.

Relation to the AFRY figure. The consultancy AFRY calculated for the government rapporteur's report that a 2,500-megawatt data centre fleet would raise the annual average price by ten per cent if consumption does not flex. I obtained 78 per cent for the same load. Neither figure is an error. AFRY's scenario builds new onshore wind in proportion to the data centres' energy need. Mine builds nothing.

The entire eightfold difference is build rate, and it is the single most important finding in this analysis.

15 What helps and what does not

I tested the counter-measures with the same model. Added load 1,600 megawatts, season 2025/26.

Table 13. Counter-measures against a 1,600-megawatt load

Scenario	Average price	Households	Elec.-heated house
Load arrives, nothing done	83.70	+546 m€	+331 €/yr
Demand flexibility: load withdraws from the most expensive hours			
2 % most expensive	83.43	+540 m€	+327 €/yr
5 % most expensive	82.59	+516 m€	+313 €/yr
20 % most expensive	77.08	+367 m€	+220 €/yr
New wind capacity			
×1.25	73.86	+344 m€	+208 €/yr
×1.50	66.25	+187 m€	+113 €/yr
×2.00	56.63	−20 m€	−14 €/yr
New weather-secure capacity			
+500 MW	75.14	+379 m€	+230 €/yr
+1,000 MW	66.80	+212 m€	+129 €/yr
+1,600 MW	56.51	±0 m€	±0 €/yr
Flexibility 5 % and wind capacity ×1.5	64.82	+145 m€	+87 €/yr

15.1 A correction to my own earlier claim

I was about to write that the problem is 278 hours and that demand flexibility is the cheap fix. The table above shows that to be wrong as far as the bill is concerned. Removing the load from the most expensive five per cent of hours cuts five per cent off the effect. Removing twenty per cent gets a third.

The reason appears when the price effect is split by price decile. At an added load of 1,600 megawatts the cheapest decile carries 7.9 per cent of the effect and the most expensive 10.4. The rest is spread evenly in between.

The bill is not made in the tail. It is made across the whole year. The load raises residual load in every hour, and the supply curve is steep in the middle range as well — not only at the peak.

Two separate conclusions follow, and they must not be conflated. **Adequacy** — whether capacity suffices at the critical hour — is answered by demand flexibility, backup power and a capacity mechanism. **The bill** is answered only by new energy, roughly megawatt for megawatt. Selling flexibility to the reader as a price solution is wrong, and I was about to do it myself.

16 The base case: what to compare against

Until now I have compared against 2025/26. That is the wrong reference point for two reasons: demand grows in any case, and production is built in any case. Both happen without a single data centre.

I therefore construct a base case for 2030, which is the year the agreement reaches full volume and the year in which Loviisa would have closed.

Demand. The growth measured in Section 12 was 397 megawatts of flat load a year. Over four years that is about 1,600 megawatts. The figure contains weather variation, so it is on the high side.

Production. Installed wind capacity stood at 9,535 megawatts on 30 June 2026. In the base case I assume 2,000 megawatts are added by 2030. This is the least well-founded single assumption in the analysis — I could not verify the exact amount under construction from a public source — so I present the sensitivity separately.

Table 14. Sensitivity of the base case to new wind, demand +1,600 MW
The right-hand column is the annual bill of a detached house heated by electricity

New wind capacity	Average price	Weighted for that house	Annual bill
+0 MW	83.70	99.39	€1,094
+1,000 MW	79.38	94.43	€1,039
+2,000 MW (*base case*)	75.34	89.85	€989
+3,000 MW	71.60	85.58	€942
+4,000 MW	68.38	81.94	€902

A four-thousand-megawatt range moves the annual bill of that house by 192 euros. That is more than any data centre effect discussed in this analysis — the build rate settles the bill more than whether Google comes or not.

16.1 Two prices, not one

From here on the tables carry two price columns, because they differ systematically.

Average price is the arithmetic mean of all hours. It is the figure used in public.

Weighted is the same price weighted by the hourly consumption of detached houses heated by electricity. It is the price that house actually pays, and it is systematically higher, because the house consumes most when it is cold and still.

In the 2025/26 season the difference was 12.78 euros per megawatt-hour. In the base case it grows to 14.51. The gap therefore widens as the system tightens — and it widens the more, the less weather-secure capacity is built.

A detached house heated by electricity consumes 11,003 kilowatt-hours a year, and its wholesale annual bill in the 2025/26 season was 762 euros. All the euro figures below are changes to that annual bill, and they concern energy — not distribution and not taxes.

17 Security of supply: is there enough capacity

Price and adequacy are separate questions, and the previous section answered only the price. Adequacy is settled in a single hour, and that hour can be defined from the data.

In the 2025/26 season the consumption peak was 15,459 megawatts, and at that moment there were 5,703 megawatts of wind, 36.9 per cent of production. That was not a tight hour.

The tight hour was another one. Residual load — consumption minus wind, that is, what must be covered by dispatchable production and imports — peaked at 14,236 megawatts. At that moment consumption was 14,399 megawatts and **Finland's entire 9,535 megawatts of installed wind produced 162 megawatts**. That is 1.7 per cent of the installed base.

Those hours are not scattered through the winter. In the 2025/26 season residual load exceeded 13,000 megawatts in 131 hours, and every one of them fell in January or February: 65 in January and 66 in February. Nine hours exceeded 14,000 megawatts, five in January and four in February. Of the hundred highest residual load hours, all one hundred fell within those two months, and the peak hour was 30 January 2026 at nine in the morning.

That 14,236 megawatts is the yardstick for this section: it is the highest residual load the system has demonstrably covered.

17.1 What building more wind brings to this hour

Still and heavily loaded hours — wind below ten per cent of production and consumption above 11,000 megawatts — numbered 709 in the season. In them consumption averaged 12,638 megawatts and wind production 487, that is 5.1 per cent of installed capacity.

From this follows a figure worth reading twice. The base case's 2,000-megawatt wind addition produces **102 megawatts** in an average still and heavily loaded hour.

In the annual balance the same 2,000 megawatts is significant: Table 14 showed it lowering the annual bill of a detached house heated by electricity by 105 euros. At the critical hour it is of the order of a hundred megawatts, less than a tenth of one Loviisa reactor.

17.2 The residual load peak in 2030

Table 15. Residual load peak by scenario
In 2025/26 a maximum of 14,236 MW was covered

Scenario	Peak	Above what was covered	Firm capacity
Realised 2025/26	14,236 MW	—	unchanged
P. Base case 2030	15,802 MW	+1,566 MW	unchanged
A. Base case and Loviisa closes	15,802 MW	+2,580 MW	−1,014 MW
B1. Base case and Google 300 MW	16,102 MW	+1,866 MW	unchanged
B2. Base case and Google 500 MW	16,302 MW	+2,066 MW	unchanged
D. Base case and the full 5,000 MW wave	20,802 MW	+6,566 MW	unchanged

The column "above what was covered" states how much new dispatchable capacity or import is needed beyond what the system has already demonstrated it can cover. Row A also includes the loss of Loviisa, which reduces available firm capacity by the same amount as the gap grows.

Three observations.

The gap arises without data centres. The base case alone takes residual load 1,566 megawatts above what has been covered. Ordinary demand growth is enough for that.

Google's share is small but not trivial. A load of 300–500 megawatts increases the gap by the same amount, because flat load does not withdraw from the critical hour. Relative to the base case's 1,566 megawatts it is a fifth to a third.

And row A is worse for security of supply than row B2. Closing Loviisa removes 1,014 megawatts of firm capacity; Google's 500-megawatt load increases the requirement by 500. The difference is 514 megawatts in the agreement's favour — the same mechanism as in the price, but here it is capacity rather than euros.

17.3 Where the gap is covered from

Imports from Sweden peaked at 3,298 megawatts during the season, and in the most expensive one per cent of hours the link was near full in one hour out of three. Section 10 showed that no significant additional volume is available from there in precisely those hours, because the neighbour has the same weather.

Three options remain: weather-secure domestic production, demand flexibility, and backup power. The latter two are covered in the policy recommendation and are the cheapest — but they do not fix the bill, as Section 15 showed. The first fixes both.

From this arises the asymmetry of this analysis, and it is worth stating out loud. Wind fixes the bill but not adequacy. Flexibility and backup power fix adequacy but not the bill. Only weather-secure production fixes both, and it is the slowest to build.

18 Batteries: what they solve and what they do not

Battery storage is regularly presented in the discussion as the solution to capacity shortage, and Google's package includes a 94-megawatt battery in Kajaani. The question is calculable, and the answer turns on one ratio: power to energy.

Finland's battery fleet stood at about 1.6 gigawatts in July 2026. Fourteen projects are under construction, totalling 995 megawatts and 2.8 gigawatt-hours. A further 11.9 gigawatts and 36 gigawatt-hours are planned.

Table 16. Power and energy of the battery fleet

Fleet	Power	Energy	Duration at full power
Kajaani battery (Google package)	94 MW	0.19 GWh	2.0 h
Finland's fleet, July 2026	1,600 MW	approx. 4 GWh	2.5 h
Fleet plus what is under construction	2,600 MW	approx. 6.8 GWh	2.6 h
All planned projects	11,900 MW	36 GWh	3.0 h

The duration is two to three hours in every case. That is not a shortcoming but a design choice: a battery earns its money from frequency reserves and intraday price arbitrage, and two hours of energy is enough for those. The problem arises only when the same equipment is offered as the answer to a different question.

18.1 How long the tight periods are

Tight periods are not hours. I computed them directly from the data: continuous periods in which residual load exceeded a given threshold, and the energy above the threshold during them.

Table 17. Continuous tight periods, 1 Aug 2023 – 31 Aug 2026

Residual load threshold	Periods	Longest period	Largest energy above threshold
11,000 MW	87	141 h (5.9 days)	213 GWh
12,000 MW	45	59 h (2.5 days)	66 GWh
13,000 MW	26	20 h (0.8 days)	16 GWh
14,000 MW	2	5 h	1 GWh

The difference between seasons is large and worth noting. In 2024/25 — mild and windy — there was not a single period longer than twelve hours in which residual load exceeded 12,000 megawatts. In 2025/26 there were thirteen, and in 2023/24 nine. One mild winter tells you nothing about the system.

The longest periods of the 2025/26 season were 29–31 January (50 hours, 65.6 GWh above the threshold) and 5–8 February (59 hours, 29.8 GWh). Wind production during them averaged 306 and 986 megawatts.

18.2 How often and how long: 86 winters

Three seasons are not enough for dimensioning. I have treated the long time series separately in my analysis *Kolmas potenssi* (19 August 2026), and its results belong alongside these, because they change the picture in two directions.

The data are six ERA5 grid points (Tankar, Kristiinankaupunki, Pori/Tahkoluoto, Kemi/Ajos, Kouvola, Ilmajoki) and 86 winters from 1941 to 2026. A regional event requires the condition to be met at five points out of six on the same day.

Table 18. Risk days and periods in 86 winters, regional criterion 5/6

Criterion (10 m, daily mean)	Days per winter	Longest period	At least 8 days
< 3 m/s and $< -10\,°$C (extreme)	2.5	5 days	0
< 4 m/s and $< -10\,°$C (corrected, lower)	7.7	12 days	6
< 5 m/s and $< -10\,°$C (corrected, upper)	11.5	15 days	20

The choice of criterion matters and is argued separately: the three-metre threshold rests on turbine cut-in speed, but considerably less stillness suffices to break the balance — the arithmetic of the power curve and a sensitivity analysis over 27 combinations both give a threshold of 4 to 5 metres per second at ten metres' height.

For the next ten years this means 77–115 risk days and **one to two periods of at least eight days**. The longest possible continuous period given by the data is 12–15 days.

It is worth returning here to the previous subsection. The longest period I measured in three years of data was 59 hours, or 2.5 days. The design case is 12–15 days. **A three-year sample does not contain the tail, so the coverage percentages in Table 19 are upper bounds** — in the actual design case they are a fraction of what is shown.

18.3 Does warming make the problem go away

This is the most common counter-argument, and there is an 86-year answer to it.

The number of risk days per winter shows no statistically significant trend, even though the average winter temperature has risen clearly over the same period. The phenomenon is neither becoming more common nor rarer.

Two things follow. The risk can be dimensioned, because a stable frequency makes the historical record a usable basis for dimensioning — an investment or a reserve requirement need not rest on a forecast but on a recurrence rate. And the change is elsewhere: in January 2026 wind conditions on the coast were exceptionally weak but not unprecedented (z-score -1.48) and the temperature was ordinary (-0.97), yet the period was among the most severe in the entire record in terms of system impact. The cause was not the weather but the fact that consumption was above 15 gigawatts where the same weather previously met 13 to 14.

The weather recurs at the same frequency as before. What grows is the consequence of that same weather.

Two qualifications belong here. The phenomenon is mostly local: 45 per cent of all risk days concern only one grid point out of six and only 2.6 per cent concern all six, so geographical diversification works. But **inland points are clearly the most exposed** (12–13 days per winter on the strict criterion) and the offshore point the least (2 days) — so wind capacity moving inland increases exposure systematically.

And finally what cannot be forecast. No connection was found between El Niño (ONI) and risk days in three independent tests, because ONI and NAO are uncorrelated in this data ($r = -0.02, p = 0.73$). NAO itself does separate the risk: 8.4 risk days in the lowest tertile against 3.9 in the highest, with non-overlapping confidence intervals. Next winter's risk therefore cannot be forecast — but it can be dimensioned.

18.4 How much a battery would cover

I combine these two. The assumption is favourable to the battery: it is full when the period begins, and there is no opportunity to recharge during it — which matches reality, because a tight period is by definition one in which no surplus electricity exists.

Table 19. What share of a period batteries would cover

Battery fleet	59 h / 29.8 GWh	50 h / 65.6 GWh	141 h / 213 GWh
Kajaani battery	0.6 %	0.3 %	0.1 %
Finland's fleet, July 2026	13.4 %	6.1 %	1.9 %
Fleet plus what is under construction	22.8 %	10.4 %	3.2 %
All planned, 11.9 GW	100 %	54.9 %	16.9 %

The last row is the one to read twice. **If every battery project planned for Finland were built — 11.9 gigawatts and 36 gigawatt-hours — it would cover a little over half of one January period and a sixth of the longest.** After that it would be empty, and it could not be recharged before the wind returned.

The Kajaani battery covers 0.3 per cent of the most energy-intensive period. That is not a criticism of the battery — it is correctly sized for the task it was built for, namely smoothing the data centre's own consumption and providing frequency reserve. It is a criticism of the claim that batteries solve the capacity shortage.

A battery moves electricity between hours within the same day. A capacity shortage lasts days. They are different problems, and one is not solved by more of the other but only by a different kind.

19 N-1: what if a large unit trips

Everything above assumes the production plants are running. The power system is not dimensioned on that assumption.

Fingrid's dimensioning fault is Olkiluoto 3, rated at 1,600 megawatts. A system protection scheme has been built for it: when the plant's breaker opens, pre-selected consumption sites in southern Finland disconnect load within half a second. For the period 1 July 2025 to 31 December 2026, 530 megawatts of capacity from nine companies has been selected, the daily requirement is 350 megawatts, and availability is paid at a capacity-weighted average of 8.0 euros per megawatt per hour.

The system protection scheme solves frequency. It does not solve the capacity balance for the following days.

19.1 How often this has happened

This can be measured. I went through nuclear production hour by hour from 1 August 2023 to 31 August 2026 and searched for events in which output fell by more than 800 megawatts within three hours.

There were **eleven events in three years**, about four a year. The largest single drop was 1,628 megawatts on 19 November 2023. Some recovered within the same day, some did not: in June 2024 output stayed below 90 per cent of maximum for 19 days and in September 2024 for nearly 32.

In winter months nuclear has averaged 90 per cent of its observed maximum and in other months 78 per cent — annual outages are scheduled for the summer, and rightly so. But **15.7**

per cent of winter hours were hours in which less than 80 per cent of maximum nuclear capacity was available. That is not rare.

At the hundred tightest hours the system has nonetheless been lucky. Nuclear availability was 94–99 per cent of maximum in all three seasons. Even so, among the hundred tightest hours of the 2024/25 season there is one hour in which nuclear output was 2,623 megawatts, nearly 1,800 short.

Three seasons are not enough to estimate a probability. They are enough to establish that the coincidence is not impossible — there are well over a hundred tight hours in a winter and a nuclear shortfall in 15.7 per cent of winter hours.

19.2 What it would cost

Table 20. Loss of the largest unit on top of the base case
Annual bill of a detached house heated by electricity, energy excluding distribution and taxes

Scenario	Avg price	Weighted	Annual bill	vs today
Realised 2025/26	56.51	69.29	€762	—
Base case 2030	75.34	89.85	€989	+226
Base case and Google 500 MW	83.74	98.62	€1,085	+323
Base case and OL3 unavailable	104.90	119.89	€1,319	+557
Base case, Google and OL3 unavailable	115.51	129.79	€1,428	+666
Base case, Loviisa closed and OL3 unavailable	127.32	140.28	€1,544	+781

The loss of one large unit costs a detached house heated by electricity 334 euros a year on top of the base case. That is three times Google's own effect.

And the last row shows why the comparison in Section 21 is not academic. In a world where Loviisa has been closed and Olkiluoto 3 is in outage or has failed in January, the same house pays 781 euros a year more than today — its annual bill doubles. An agreement that keeps Loviisa on the grid is at the same time an insurance policy against this.

Is security of supply assured. The answer is conditional. On frequency, yes: the system protection scheme, inertia and frequency reserves are dimensioned to the dimensioning fault. On the capacity balance, not fully: the residual load peak is already 1,566 megawatts above what the system has demonstrably covered in the base case alone, and the loss of the largest unit would double that gap. Batteries do not provide the answer, because they cover hours and the gap lasts days.

20 Phasing out peat: what it cost

The previous section showed that adequacy is settled by weather-secure capacity. Over the past ten years Finland has made one large change to precisely that capacity, and it is worth calculating out — not least because it is debated without numbers.

Peat use in energy peaked in the 2000s at nearly 30 terawatt-hours a year. In 2015 it was about 16 terawatt-hours and in 2024 only 1.5 per cent of total energy consumption, or roughly 5.4 terawatt-hours. The fall is about two thirds in ten years.

The cause is not mainly domestic. The emission allowance price rose from five euros to over thirty between 2017 and 2020, and Matti Kahra of the Confederation of Finnish Industries stated as early as 2021 that peat's crisis stemmed from European emissions trading rather than from domestic decisions. Finland's own additions were the peat tax and one older classification decision, to which I return.

20.1 What peat did at the critical hour

Peat was burned mostly in municipal and industrial CHP plants. Their significance at the critical hour can be measured from Fingrid's real-time data.

Table 21. Balance of the hundred tightest hours, season 2025/26

Item	MW	Share of residual load
Residual load	13,525	—
Nuclear	4,226	31.2 %
Net imports	2,837	21.0 %
District heating CHP	**2,386**	**17.6 %**

Those 2,386 megawatts are the block that peat used to fuel. And here is an observation that inverts the usual framing.

The capacity did not disappear. The fuel changed. District heating CHP at the tightest hours averaged 2,113 megawatts in the 2023/24 season, 1,900 in 2024/25 and 2,386 in 2025/26. The plants did not leave; they were converted mainly to wood chips.

The security-of-supply effect of phasing out peat is therefore not measured in megawatts. It lies in whether the fuel is there in a crisis and what it costs.

20.2 What capacity that does not run costs

If CHP does not run — because fuel is expensive or unavailable — the effect is calculated with the same model as the loss of Loviisa.

Table 22. Loss of weather-secure capacity, season 2025/26
Annual bill of a detached house heated by electricity, energy only

Capacity lost	Avg price	Weighted	Annual bill	Change
None	56.51	69.29	€762	—
−200 MW	59.99	73.28	€806	+44
−400 MW	63.41	77.15	€849	+87
−600 MW	66.80	80.98	€891	+129
−1,000 MW	73.45	88.31	€972	+209

Every hundred megawatts that does not run at a tight hour costs a detached house heated by electricity about 22 euros a year.

20.3 Security of supply

The argument made by the bioenergy association is structurally valid: the supply security of fuel peat can only be assured if peat is competitive under normal conditions. If commercial use falls to zero, the contractors, the machinery and the logistics disappear — and the emergency stockpile cannot be filled even if the law requires it. A stockpile is not a separate system but a by-product of a functioning supply chain.

The emergency stockpiles hold about three terawatt-hours. At current use that is a good six months; at the 2015 level, two months.

The replacement fuel is wood chips. Replacing peat entirely would require more than eight million solid cubic metres of wood, and about a tenth of Finland's wood use has been imported — from Russia until 2022. Domestic peat was therefore partly replaced by imported wood in precisely the years when imports from the east ended.

And the fuels behave differently in storage. Peat keeps for years. Chips compost, heat up and lose energy content within months. A security stockpile that must be built from chips is a different thing from one built from peat.

20.4 Climate

Here the decision is at its strongest. Peat's emission factor is higher than coal's, and VTT has found peat's greenhouse impact to be worse than coal's regardless of whether the peat was taken from a natural or a drained mire. The emission reduction from phasing out has been estimated at about eight million tonnes of carbon dioxide.

Three counterweights are worth recording.

The waterbed effect of emissions trading: allowances freed in Finland are available elsewhere in Europe, so the effect on total emissions is not as large as the plant's own reduction. The market stability reserve cancels part but not all of it.

The carbon debt of wood chips is counted as zero in emissions trading. In the atmosphere it is not zero. This is a scientifically contested question and I do not settle it here.

And area: energy peat production is a fraction of drained peatlands, and the bulk of peatland emissions comes from drained forest and agricultural land rather than from peat extraction. Ending energy peat therefore does not touch the part from which the emissions mainly come.

20.5 Sweden did not make the same decision

Finland classified peat as a non-renewable energy source in 2000. In Sweden peat still counts among renewables and is eligible for electricity certificates in power generation — although environmental permit conditions have tightened and an estimated one application in ten is approved.

The consequence is one nobody planned. Sweden now exports peat to Finland's emergency stockpiles.

Finland therefore made a unilateral classification decision that raised the cost of its own domestic fuel, and now buys the same fuel from a neighbour that did not make that decision. For the atmosphere the difference is zero.

20.6 Assessment

Three decisions should be separated, because they are not equally good.

Emissions trading did the work, and that was right. Peat's retreat was market-driven and would have happened without domestic measures. It is not open to criticism.

The peat tax was overlapping steering. It accelerated a change already under way and removed the supply chain faster than the security-of-supply arrangement could adapt. The problem was the timing, not the direction.

The 2000 classification decision is the one to regret. It placed Finland in a different position from Sweden without a corresponding climate benefit, and the outcome is peat imported from where it is renewable.

The price and capacity effect is nonetheless smaller than the debate suggests, because the capacity is still on the grid. This is not a question of lost megawatts but of fuel risk and cost — and that risk is real, because 17.6 per cent of the critical hour's residual load sits in plants whose fuel supply has been rebuilt from scratch in ten years.

21 The net effect of the agreement

Table 23. Scenarios in 2030: annual bill of a detached house heated by electricity
Energy share of the bill; distribution and taxes not included
Base case: demand +1,600 MW and wind capacity +2,000 MW

Scenario	Avg	Weighted	€/yr	vs today	vs base
Realised 2025/26	56.50	69.28	762	—	—
P. Base case 2030, no Google	75.34	89.85	989	+226	—
A. Base case and Loviisa closes	93.41	108.61	1,195	+433	+206
B1. Base case and Google 300 MW	80.29	95.02	1,046	+283	+57
B2. Base case and Google 500 MW	83.74	98.62	1,085	+323	+97
C. Base case and Google 1,600 MW	104.90	119.89	1,319	+557	+331
D. Base case and the full 5,000 MW wave	175.46	179.72	1,977	+1,215	+989

The first figure is the most important, and it has nothing to do with Google. The wholesale annual bill of a detached house heated by electricity rises by 226 euros without a single data centre, purely on ordinary demand growth. That is 30 per cent of the current annual bill. Everything else in this table is on top of that.

Google's own share is 57–97 euros a year on the best load estimate, in the same annual bill.

And the agreement's net effect is still clearly positive. Row A is the world in which no agreement is made and Loviisa closes in 2030: it costs 206 euros a year on top of the base case. Rows B1 and B2 cost 57–97. The difference is 109–149 euros a year.

Here it is worth being precise, because in earlier versions I phrased it wrongly. **The bill does not fall in any scenario.** It rises in all of them. The agreement means it *rises 109–149 euros less* than it would have risen without the agreement.

The reason is mechanical and worth saying out loud. The price is set by residual load, that is consumption minus wind. Closing Loviisa raises residual load by 1,014 megawatts; Google's load raises it by 300–500. The difference is 514–714 megawatts in the agreement's favour — the agreement does not bring electricity, but it prevents more from leaving than it adds in consumption.

Row D is the one to fear. The full data centre wave on top of the base case is 989 euros a year, tripling the house's wholesale electricity bill. Google's agreement is a small part of it and the only part that brings a counterweight.

21.1 The best estimate of Google's load

The break-even is known. The load is not, but there is now a substantially better estimate of it than on the day of the announcement.

Jussi Kangasharju, professor of computer science at the University of Helsinki, who has studied data centre projects for fifteen years, has published a calculation. He was present at Google's event, and at a side session held under the Chatham House rule it was stated that the 13 billion figure includes the computers. Computers typically account for 70–80 per cent of an investment's value, that is about ten billion. Some three billion is left for buildings and building services. The industry rule of thumb is that a billion euros corresponds to about a hundred megawatts of capacity — or, put the other way, that the construction cost of a hundred megawatts without servers is a billion.

From this follows **300–500 megawatts**.

The same result by another route: 13 billion is just under three times the 4.5 billion Google has said it invested in Hamina, and general estimates of Hamina's capacity range from 100 to 250 megawatts. For the new investments that gives 300–750 megawatts.

This is three to five times smaller than the estimate offered on the day of the announcement, and it is better founded: it rests on two independent rules of thumb and on one confirmed fact about the content of the investment sum.

I re-ran the simulation over this range.

Table 24. Net effect of the agreement as a function of Google's load
Net = with the agreement minus row A (no agreement, Loviisa closes)

Google's load	Avg price	Households	Elec.-heated house	**Net**
300 MW · Kangasharju, lower	61.70	+108 m€	+65 €/yr	−**147** €/yr
400 MW	63.41	+143 m€	+87 €/yr	−**126** €/yr
500 MW · Kangasharju, upper	65.11	+178 m€	+108 €/yr	−**104** €/yr
750 MW · upper end via Hamina	69.30	+262 m€	+159 €/yr	−53 €/yr
1,014 MW · break-even	73.68	+350 m€	+212 €/yr	±0
1,600 MW · estimate on the news day	83.70	+546 m€	+331 €/yr	+119 €/yr

On the best available estimate the agreement benefits the consumer: the annual bill of a detached house heated by electricity rises 109–149 euros less than in the world where no agreement was made and Loviisa closed.

One addition points the same way. The average utilisation rate of data centres is around 60 per cent according to industry calculations — such systems are never run close to full capacity, because a swing in load would then break service agreements. My simulation assumes flat load at nameplate capacity. If 300–500 megawatts is the nameplate and average load is 60 per cent of it, the benefit of the agreement grows to 147–173 euros a year.

The number is therefore still not disclosed, and it remains the title of this analysis. But it is no longer unknown in the sense it was on the day of the announcement, and the best estimate places it clearly below the break-even.

21.2 Why the number was not given

Here is a question that deserves to be asked. If the load is 300–500 megawatts, it is small relative to what was feared — about a thirtieth of Finland's generating capacity. Disclosing it would have calmed exactly the concern that arose after the announcement. Why did the company not disclose it?

I mark this as theory. I do not know the motives, and what follows is inference from what is publicly visible.

The number would be a ceiling. The announced investment covers two years, but the sites reserved are larger and the Hamina centre grew to its present size over thirteen years. Publishing four hundred megawatts today would make a later expansion look like a broken promise. Saying nothing preserves room to move.

The number would be a negotiating position. Disclosing the load tells electricity sellers, the transmission operator and municipalities how much the buyer needs. That is not information given away free before the contracts for the next phases.

And the number would have shrunk the news. This is the most interesting of the three, because it also explains what else was withheld. Beside thirteen billion, four hundred megawatts leads straight to the question Kangasharju asked: if the capacity is that size, where does the rest of the money go? The answer is that about ten billion is computers, which are not manufactured in Finland. The company declined to break the investment sum down by category when asked.

Two numbers therefore tell different stories. The large sum says a great deal is being invested in Finland. The small capacity says little of that sum stays here. Both are true, and only one was published.

From this follows an outcome that is perverse from the company's point of view. Silence protected the investment story but created the electricity concern that the number would have dispelled. A Finn fears the electricity bill, not how many billions go abroad. The break-even used in this analysis resolves in Google's favour — but it was resolved by an outside professor's calculation, not by the company's own information, and only three days after the news.

21.3 Counter-argument: was the closure a real threat

The whole result of this section rests on row A, that is, on Loviisa having been closed without the agreement. The claim is Fortum's own, and it was made in a situation where making it benefited the company. It deserves its counter-argument, and one has been made publicly.

Juha-Matti Mäntylä of Yle has argued that Loviisa's fate was hardly genuinely in the balance: the government granted the plant an operating licence to the end of 2050 already in 2023, and the extension had been in preparation for years.

An operating licence and an investment decision are, however, different things, and the difference is decisive here. An operating licence states that the state permits the plant to run. It does not state whether the owner will finance it. Fortum's lifetime extension programme is about one billion euros, and 80 per cent of the projects and 700 million euros were without an investment decision on the eve of the announcement. The licence had therefore been in force for three years without a financing decision being made.

A third voice came from the owner. Minister Joakim Strand, responsible for state ownership steering, who said he heard about the project only in the final stages of preparation, stated that the agreement ensures Fortum carries out the substantial investments required by Loviisa's lifetime extension. The statement supports Fortum's reading, but its weight is limited by the fact that the state owns more than half of the company and the minister is therefore not an independent observer.

Which reading is correct cannot be settled from outside. A company always has an interest in presenting an investment as more necessary than it is, and Mäntylä's doubt is justified in that respect. But the claim *the licence existed, therefore the plant would have continued* does not follow, because a licence does not pay for a modernisation.

I record this as uncertainty and do not resolve it. The practical consequence is clear: if the closure was not a real alternative, the correct comparison is today's situation rather than row A, and the agreement's net effect is then Google's whole load — row C as it stands, 331 euros a year for a detached house heated by electricity rather than 119. The counter-argument therefore worsens the assessment of the agreement from the consumer's point of view rather than improving it.

22 Does this change the conditions for new nuclear

Section 15 concluded that only weather-secure capacity fixes both the bill and adequacy, and Section 17 that building more wind brings almost nothing to the critical hour. From that follows a question that has been unanswered in Finland for fifteen years.

22.1 Two licences that do not exist

It is repeated in public that Finland has two unused nuclear licences. The formulation is misleading, and the precision matters.

Olkiluoto 4 received a decision in principle confirmed by Parliament in July 2010. The condition was that a construction licence had to be applied for by June 2015. In May 2014 TVO requested a five-year extension because Olkiluoto 3 was late; the radiation safety authority found no safety obstacle to an extension, but the government rejected the request by 10 votes to 3 in September 2014. **The decision in principle lapsed on 30 June 2015.** TVO had by then spent about 80 million euros on the project.

Fennovoima's Hanhikivi 1 received a decision in principle in 2010 and a supplementary decision approved by Parliament in 2014 by 115 votes to 74. The project ended in 2022 with Russia's invasion of Ukraine; the plant supplier and 34 per cent owner was Rosatom's RAOS.

Neither is therefore a valid licence. A new project needs a new decision in principle from the start. The correct formulation is that Finland has two plant sites with completed environmental studies and partial earthworks — not two ready licences.

22.2 What has changed: the long purchase agreement as a financing instrument

Here is what was not said on the news day and what may prove to be the agreement's most important long-term consequence.

In the United States, data centre companies have signed thirteen nuclear agreements in two years, totalling more than 9.8 gigawatts. Microsoft committed through a 20-year agreement to restarting the Three Mile Island unit (835 MW), Meta to a 20-year agreement on the Clinton plant (1,121 MW), Amazon to up to 1,920 megawatts through to 2042, and Google ordered 500 megawatts of small reactors from Kairos Power.

The agreements fall into two kinds, and the difference is decisive. Some *preserve* existing capacity — restarting it or preventing closure. Some *finance new build*, such as Google's agreement with Kairos or Amazon's 700-million-euro investment in X-energy.

Google's agreement with Fortum belongs to the first kind. It preserves Loviisa; it does not finance a new reactor. In that sense it does not directly advance the building of new nuclear.

But it does something else, and Energiateollisuus's electricity market director Pekka Salomaa said it on the day of the announcement: no purchase agreement covering the output of a nuclear plant has been made in Finland before. In Fortum's view it is the largest such agreement ever in Europe.

The obstacle to new nuclear in Finland has not been technology and it has not been permits. It has been that nobody has been willing to commit to buying electricity twenty years in advance. The Mankala model solved that historically through ownership: a shareholder pays cost price and carries the risk. A long purchase agreement solves the same thing without ownership — and Finland now has a first precedent for it, priced and signed.

It does not build a reactor. It demonstrates that there is a buyer in Finland too who will sign for 22 years.

22.3 What construction actually takes

Olkiluoto 3 is a poor sample. Construction began in August 2005 and regular production in April 2023 — more than sixteen years, and it was the world's first EPR.

A broader sample gives a different picture. Fifty nuclear plants whose construction began after Olkiluoto 3 were completed in an average of six years and five months. South Korea's APR1400 units have been built in 52 months, and plants based on China's own designs have been completed in about five years.

Construction is therefore not what takes time in Finland. What precedes it is.

Table 25. Timetable for a new large unit if the decision were taken now

Stage	Estimate
Application for a decision in principle under the new act	2027 at the earliest
Government decision in principle	1–2 years
Supplier tendering and construction licence	2–3 years
Construction (international average 6 yr 5 mo)	5–8 years
On the grid at the earliest	**2036–2040**

One thing follows from this that matters for this analysis. **New large nuclear does not affect the base case of Section 16 at all**, because the base case runs to 2030. It affects only what happens after that — and that is precisely why the decision should be taken now rather than when the gap has arrived.

22.4 The new Nuclear Energy Act

The government submitted a bill for a new Nuclear Energy Act to Parliament on 12 March 2026. It would repeal the 1988 act and amend fourteen others; the bill runs to more than 550 pages. The act is intended to enter into force on 1 January 2027, which would allow a decision in principle to be applied for under the revised rules still within this parliamentary term.

Three changes matter.

Decision-making moves from Parliament to the Government, and for the smallest projects to the Ministry of Economic Affairs and Employment; the Government's decision is submitted to Parliament as a report. The licensing model becomes modular, so that if a project changes along the way, not all permits and statements need to be applied for again. And the radiation safety authority would license small nuclear facilities directly.

Heikki Vestman of the National Coalition has called the reform the single most significant energy policy decision of the decade. It does not remove the dimensioning obstacle — the political uncertainty of a decision in principle — but it removes one: a project no longer falls on a single parliamentary vote.

22.5 Small modular reactors

A small reactor produces at most 300 megawatts and is intended for series production. In Finland, Steady Energy, which originated at VTT, is developing the LDR-50 reactor, and **construction of the first pilot plant began in Helsinki in February 2026.**

Here is a point worth reading carefully, because it connects to the observation in Section 12.

The LDR-50 does not produce electricity. It produces district heat. It therefore adds no megawatts to the power balance — but it *removes* load, and it removes it from precisely where the measured demand growth comes from. In Section 12 the consumption of utilities and energy supply grew 37 per cent in a year while metering points grew 1.4 per cent, and the reason is electric boilers in district heating networks. According to Fingrid, the combined capacity of electric boiler projects in operation and under construction exceeded three gigawatts in June 2026.

For residual load, a gigawatt of removed electric boiler load is the same thing as a gigawatt of new weather-secure production. A small reactor placed in district heating is therefore a production plant as far as the capacity balance is concerned, even though it feeds nothing into the grid.

A qualification is due. There are dozens of small reactor concepts worldwide, and only a few have reached construction or completion. The cost advantage of series production is a promise, not a realised fact, and no concept yet has a series. The completion and operating experience of the first Finnish pilot will settle whether this is a solution for the 2030s or the 2040s.

22.6 What the state can do

The permitting side is now handled. What is not handled is risk.

Nuclear's problem is not its production cost but the fact that the entire cost is paid up front while the return arrives over forty years at a market price nobody can forecast. Honkapuro's observation therefore holds: investors have not found the projects profitable — not because production is expensive, but because the revenue cannot be promised.

Three instruments would change that, and they are of different sizes.

Enabling and standardising long purchase agreements. The cheapest, and already begun. Google's agreement with Fortum is a precedent; if similar ones emerge for new projects, the state is not needed as a financier at all. The state can accelerate this by ensuring that competition and electricity market regulation clearly permit twenty-year contracts.

A contract for difference. The state guarantees a strike price and recovers the difference if the market price exceeds it. In use in the United Kingdom. Not a direct subsidy but a transfer of risk, and it costs something only if the price falls below the agreed level.

A capacity mechanism pays for availability rather than energy, and it is already item 6 in my policy recommendation. It improves the economics of all weather-secure plants under the same rule and does not pick a technology.

I do not recommend one of these, because the choice is political rather than computational. What I do state is what is computational: without one of them no new large nuclear will be built, and that has been known since Olkiluoto 4 lapsed, that is for eleven years.

23 Investment versus bill

A separate question is whether the investment that stays in Finland offsets the bill.

Jouni Salonen, senior adviser at Business Finland, estimates that 20–30 per cent of the billions in data centre investments ends up in Finland; the rest is machinery and equipment not bought here. Of the construction workforce, 50–70 per cent is estimated to come from Finland. The Ministry of Finance has estimated that about a quarter of Google's investment stays in Finland. Kangasharju's breakdown sets an upper bound on this: if computers are 70–80 per cent of the sum, at most about three billion, or 23 per cent, is left for the share staying in Finland — and part of that goes to foreign contractors. The bottom row of the table is therefore the most realistic and the top row too high.

The investment is one-off; the bill recurs every year. The comparison must be made in present value. At a discount rate of four per cent and the agreement's 22-year horizon the annuity factor is 14.45.

Table 26. Break-even: the annual additional bill that offsets the investment

Share staying in Finland	Annual cost	Apartment	Elec.-heated house
20 % = €2,600 m	€180 m/yr	€4/yr	€27/yr
30 % = €3,900 m	€270 m/yr	€6/yr	€41/yr
33 % = €4,333 m	€300 m/yr	€7/yr	€45/yr

According to the Energy Authority, household electricity bills fell by an average of 10–20 euros in 2025 and by an average of 75 euros in 2024. The break-even is therefore smaller than the ordinary year-to-year variation in an electricity bill. The whole share of the investment that stays in Finland disappears into a price change nobody would notice on their bill.

And compared with the net effect in Section 21: on the best load estimate the net is negative, that is, the bill does not rise but falls relative to the alternative. In that case the break-even is not the relevant question at all — the consumer gets both the investment and cheaper electricity. It would turn only if the load proved larger than 1,014 megawatts: at 1,600 megawatts an additional 119 euros a year would exceed the break-even on all three share assumptions.

24 Public finances

From 1 July 2026 the electricity tax is 2.24 cents per kilowatt-hour with a security-of-supply levy of 0.085, that is 2.325 in total. From a flat load that is 92 million euros a year at five hundred megawatts and 293 million at 1,600. The compensating support fell in the budget session, so it remains in full.

For comparison: Finland's five largest data centre operators paid a total of about 105 million euros in corporate and property tax over 2019–2023, that is about 21 million a year. In the other direction: the largest single value-added tax refund to one operator was 240 million euros in one year, because data centres may deduct the VAT on their purchases without selling a VAT-liable service in Finland. The Director General of the Tax Administration has stated that the net effect on public finances may be negative.

The state's ledger is positive as far as this agreement is concerned. That does not mean the country benefits evenly: the electricity tax goes to the state, while the price increase is paid from household bills and ends up with electricity producers. Both money flows run upward from the consumer.

25 Employment

Google estimates the employment effect at 37,000 people during construction and 7,000 permanent jobs. The methodology has not been published.

A measured comparison: all Finnish data centre projects together are estimated to employ about 10,000 workers at present, of whom 50–70 per cent come from Finland. The operating staff of a hundred-megawatt facility has been estimated at 64–150 people. At a load of 1,600 megawatts that would be 1,000–2,400 direct jobs.

I do not estimate indirect effects. They rest on models that are sensitive to their inputs, and the input in this case is precisely the number that was not given.

26 Changes to earlier versions

Version 2.0 was published on 9 September 2026 and has been in circulation. Versions 2.1 to 2.8 followed on 11–12 September 2026.

Version 2.8 adds Section 22 on the conditions for new nuclear: the legal status of the two "unused licences", the long purchase agreement as a financing instrument, actual construction times, the new Nuclear Energy Act and small modular reactors. It also corrects a formulation repeated in earlier versions: Olkiluoto 4's decision in principle lapsed on 30 June 2015 and is not in force, and Fennovoima's project ended in 2022 — Finland therefore has not two unused licences but two plant sites.

Version 2.7 corrects an error. In version 2.6 I gave three figures for the recurrence of tight periods (34, 5 and 0.6 times per decade) citing an unpublished weather run. The figures were wrong. The correct ones are in my published analysis *Kolmas potenssi* and have now been taken from there: 7.7–11.5 risk days per winter, a longest period of 12–15 days, and 6–20 periods of at least eight days in 86 winters. An answer to the warming argument was added at the same time: there is no significant trend in the frequency of risk days over 86 years.

Version 2.6 added Sections 18 and 19: the actual capability of battery storage to cover tight periods, the measured duration and energy of those periods, and the loss of the largest unit.

Version 2.5 added Section 20 on the effects of phasing out peat on price, security of supply and climate, together with a comparison to Sweden.

Version 2.4 added the section on security of supply, computed as the residual load peak, and rewrote the opening.

Version 2.3 added the base case. Earlier versions compared against 2025/26, which ignored the fact that demand grows and production is built in any case. A consumption-weighted price was added alongside the arithmetic average in all tables, and a wording error was corrected: earlier text spoke of the bill being lower when the point is that it rises less.

Version 2.2 added the best estimate of Google's load (300–500 MW), which turned the conclusion, and a section on why the number was not given.

Version 2.1 added the assessment of the project pipeline, the counter-argument on the Loviisa closure threat, the price cap treated with four arguments made by others, the Ministry of Finance estimate, and a note that part of the price rise is the signal needed to build new production.

27 Limitations

1. **Google's load is still undisclosed.** The 300–500 megawatt range used in Section 21.1 is an outside estimate, not information. It rests on industry rules of thumb and on one confirmed fact about the investment sum including computers. If the assumption of a 70–80 per cent computer share is wrong, the range shifts. The sign of the whole result depends on whether the load is below or above 1,014 megawatts.

2. **The load is assumed flat at nameplate capacity.** Industry figures put the average utilisation of data centres at about 60 per cent, so realised consumption is below nameplate. The simulation therefore overstates the effect, and the estimate is deliberately cautious from the consumer's point of view. At the critical hour the difference is smaller than in the annual average.

3. **The simulation holds supply fixed.** This is a deliberate upper bound, not a forecast. In reality new production is built, and Sections 6 and 15 measure how much is needed. The gap between AFRY's ten per cent and this analysis's 78 is build rate, and neither end is a forecast. A related counter-argument is worth recording: the roughly five-cent consumer price of recent years was too low for new production to be worth building. Part of the price

rise is therefore the signal whose absence is the reason production has not been built, and that part should not be read as a pure loss to the consumer. The argument is Anni Lassila's (HS Visio), and it is correct.

4. **The base case's wind assumption is unverified.** Section 16 assumes 2,000 megawatts of added wind capacity by 2030. I could not obtain the exact amount under construction from a public source — the industry project list is a paid product. The sensitivity in Table 14 shows that the range of the assumption moves the house's annual bill by 192 euros, which is more than Google's own effect. This is the single largest uncertainty in the analysis.

5. **The model does not include the Nordic response.** A Finnish price rise would increase imports and affect neighbouring prices. This overstates the effect, probably substantially at large loads.

6. **Distribution network consumption is 52.79 TWh against 84.7 for the whole country.** The difference is industry connected to the transmission grid, whose hourly profile is not in these datasets. Its exposure is assumed neutral, which probably understates the household share.

7. **Datasets 358 and 360 net out small-scale production** at the metering point, so they must not be summed with dataset 124.

8. **The series begins on 1 August 2023.** There are three seasons. Comparison with the energy crisis years is not possible with this data.

9. **Was the closure threat genuine.** Treated in Section 21.3. The claim is the company's own, it has been disputed in public, and I do not resolve it — but if the closure was not a real alternative, the assessment of the agreement from the consumer's point of view worsens rather than improves.

10. **The realisation rate of data centres.** Treated in Section 6. The connection agreement figure is the planned scope reported by project developers, not built capacity, and the realisation rate in every investment class is below one hundred per cent. The five-gigawatt rows in this analysis are upper bounds, not forecasts.

11. **Icing.** In January 2026 wind production fell to 0–5 per cent of capacity, and the cause was not stillness but ice on the blades. I have not separated how much of the low wind share during expensive hours is which. The difference matters, because icing occurs near zero degrees, that is in the temperature band that is growing.

12. **The news sources are same-day reports.** The two most important quotations are journalists' renderings rather than original recordings.

28 Sources

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[5] MTV Uutiset, 9 Sept 2026: Orpo on an electricity price cap and on keeping Meri-Pori in reserve.

[6] Uutissuomalainen, 9 Sept 2026: Antti Järvinen (Google) on data centre flexibility.

[7] MTV Uutiset, 9 Sept 2026: Iivo Vehviläinen (Aalto University) on the scale of the new demand.

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[18] Asta Sihvonen-Punkka (Fingrid), Aug 2026: estimate of consumption growth of 40 per cent in five years.

[19] Fingrid open data, datasets 358, 360, 124, 75, 188, 194, 201, 60 and 61. Interface `data.fingrid.fi/api`.

[20] Sähkötin (`sahkotin.fi/prices`): Finland's area price by hour, 1 Aug 2023–31 Aug 2026 for the simulation and 1 Jan 2015–31 Aug 2026 for the reservoir comparison, 102,264 hourly observations in total.

[21] Energy-Charts (`api.energy-charts.info`): the same series for cross-checking, and the SE3 area price.

[22] NVE, Magasinstatistikk: Norwegian reservoirs weekly 1995–2026.

[23] Statistics Finland: total energy consumption 2025; electricity prices by consumer type; peat use 2024.

[24] Energiateollisuus: Energy Year 2025, Electricity. Preliminary data, 21 Jan 2026.

[25] Energy Authority: development of household electricity bills 2024 and 2025.

[26] J. Jukola: *Kallis talvi, kallis tunti*, v. 1.0, 20 Aug 2026 — the analysis whose reservoir correlation is corrected in Section 11.

[27] J. Jukola: *Kolmas potenssi*, v. 1.0, 19 Aug 2026. Six ERA5 grid points, 86 winters 1941–2026, regional criterion 5/6; frequency of risk days, duration of periods, trend test, NAO and ENSO analysis. The long time series figures in Section 18 are from it.

[28] Suomen uusiutuvat ry, 9 Sept 2026: battery storage project statistics.

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Sources and permissions. News information comes from the companies' own releases and from news outlets reporting them on 9 September 2026. Statistical information is public and verifiable. The simulation rests on a panel of 27,002 hours whose structure and sources are described in Section 4; **the data and the calculations are available on request, including and especially for criticism**. The method is described in Section 5 in sufficient detail to be reproduced. The information, tables and conclusions in this analysis may be used, quoted and built upon freely. A source reference is a courtesy, not a condition. Sections 11, 15.1 and 18.3 correct the author's own earlier views; corrections of errors are welcome at `jujukol@gmail.com`.